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DIAGNOSTICS

# Investigation of Low Energy Electron Irradiated $SiO_2$ Based MOS Devices by Capacitance-Voltage and Thermally Stimulated Current Techniques

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**Abstract**—Silicon oxide based and aluminum gated MOS structures fabricated on n-type silicon are investigated after irradiation with low energy electrons in scanning electron microscope. The thermally stimulated current (TSC) technique in the temperature range from 80 K to 320 K revealed a number of the electron beam induced charge traps. With the help of the capacitance-voltage method, the traps revealed by the TSC were identified by their location (within dielectric, semiconductor or at the interface) and by their nature (trap for electrons or for holes).

**Keywords:** silicon oxide, MOS capacitor, thermally stimulated current, charge traps

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## 1. INTRODUCTION

For decades, silicon oxide has successfully served as a major gate dielectric for metal-oxide-silicon field effect transistors (MOSFETs) following the Moor's law until the technology reached 45 nm [1]. Up to now silicon oxide is used as a gate dielectric for silicon carbide [2], and gallium nitride [3]. Silicon oxide is also used as a passivation layer for bipolar transistor integrated circuits [4], silicon solar cells [5] and Flash memory [6], as well as an active layer for resistive memory [7].

In modern microelectronics technology electron beam lithography (EBL) has become a widely employed method for producing microelectronics and nonelectronic structures at a nano scale [8]. Irradiation with a low energy electron beam results in generation of charged traps in the silicon oxide and at the interface with semiconductor [9–11]. This can lead to degradation of characteristics and performance of the bipolar and field effect devices. The induced defects can impact the quality and integrity of insulators. The performance requirements of $SiO_2$ include stability under bias stress, radiation immunity under high and low energy irradiation, and a low density of bulk traps and states at the interface with the semiconductor.

The capacitance-voltage (C–V) method is a useful technique to monitor generation of the positive charge under e-beam irradiation [9]. However, this method is not spectroscopic and electronic properties of the generated traps cannot be determined. Thermally stimulated current (TSC) is a technique which measures emission of charge from the defect levels, whereas the temperature dependence of the TSC signal is associated with the energy levels of the generated traps [12].

The objective of this study is to investigate the properties of defects generated in $SiO_2$ based MOS capacitors by low energy e-beam irradiation with the use of two experimental methods: capacitance-voltage and thermally stimulated current techniques. The C–V measurements reveal information regarding the existence of both interface states and defects inside the oxide bulk. For deeper understanding of the origin of the defects introduced into dielectric layers and at the dielectric-semiconductor interface, we use thermally stimulated current technique. With the help of the capacitance-voltage method, the traps revealed by the TSC were identified by their location within dielectric, semiconductor, or at the interface and by their type, i.e., trap for electrons or for holes.

## 2. EXPERIMENTAL

The investigation was carried out on an oxidized n-type Si epitaxial layer with a carrier concentration of $10^{15}$ $cm^{-3}$. The oxide thickness was 45 nm, the top electrode was formed by thermal evaporation of aluminum with diameter of 1.6 mm. The charges in the dielectric and at the $Si/SiO_2$ interface were studied by the C–V technique using the Parametric Analyzer Keithley 4200A-SCS. The MOS structures were irra-

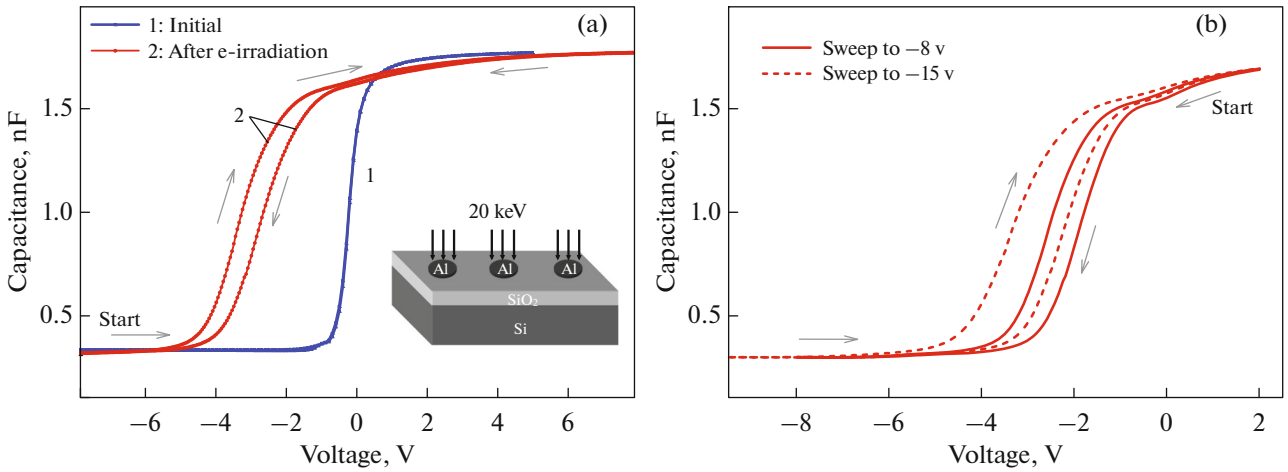


**Fig. 1.** C–V characteristics of initial and electron beam irradiated MOS structures (a) before and after irradiation; (b) irradiated MOS structure at different maximum sweep voltage to depletion. The inset presents the design of the samples and the scheme of irradiation.

diated by an electron beam with an energy of 20 keV and a dose of 500 μC/cm$^2$ in a scanning electron microscope Jeol JSM 840A with no external bias. The TSC measurements were carried out using Everbeing Cryostation (80–450 K) with 4 micromanipulators and Lake Shore temperature controller. The current was measured by Source Meter Keithley 2450. PC based system with a specially developed software was used for data acquisition and automation of the experiment.

The energy levels are filled by electrical injection at a low temperature, usually at 80 K, while during warming the captured electrons or holes are emitted and contributed to the measured current. The TSC measurements are done under various bias conditions during cooling, various polarity and magnitude of filling pulses and bias conditions during heating. The certain conditions for the bias and filling pulses are chosen based on the C–V characteristics to monitor traps either in the oxide bulk or at the Si/$SiO_2$ interface, and even in the space charge region of Si bulk. The sample is cooled down with or without bias. At 80 K a voltage pulse was applied to inject electrons or holes into the oxide bulk. During temperature raising with a constant rate under reverse bias the discharging current is monitored.

## 3. RESULTS AND DISCUSSION

### *3.1. Capacitance-Voltage Characteristics*

Fig. 1a shows the C-V characteristics of the virgin MOS capacitor and the device after electron irradiation with an energy of 20 keV and the dose of 500 uC/cm$^2$. The virgin capacitor (curve *1*) shows sharp transition from depletion to accumulation, a very low value of an oxide fixed charge and a lack of hysteresis indicating low, if any, electron and hole trap density in the oxide bulk. After electron beam irradiation (curve *2*) the whole C-V curve shifted towards a negative voltage. The slope of the curve was notably reduced, especially near accumulation, and the retrace sweep demonstrated significant hysteresis. These changes clearly imply that electron irradiation caused generation of a positive charge at the Si/$SiO_2$ interface with the highest density of the interface states localized near the edge of the conductance band. Formation of shallow hole traps in the oxide bulk is also detected. The observed changes caused by the low energy electron beam irradiation are generally similar to those previously observed in [9], with an exception of generation of deep hole traps in the oxide bulk. Indeed, as can be seen from Fig. 1b, an increase of the negative bias during voltage sweep demonstrates a shift of both forward and reverse C-V curves. Shallow traps can release their charge at room temperature as soon as the bias is off, while the deep traps can hold the charge for several seconds at room temperature. As the positive charge build-up in silicon oxide is known to complexly depend on type and energy of irradiation, method of oxidation and the electric field across the oxide during irradiation [7, 13], observed generation of deep hole traps in our devices can be attributed to oxide features caused by the oxidation conditions. To obtain information about electronic properties of the generated defects and identify their nature, we employed the method of thermally stimulated current [12].

### *3.2. Thermally Stimulated Current Spectra*

Our choice of cooling, pulsing and measurement conditions during TSC monitoring was based on the C–V characteristics presented in Fig. 1b. Figure 2a shows the TSC spectra of the electron irradiated MOS device measured under different reverse bias conditions during heating, while the cooling and pulsing conditions were the same. The TSC spectra revealed a number of the current peaks caused by release of the trapped charges. As the device is cooled down, a positive bias of 10 V is applied to drive the MOS capacitor to accumulation. The band bending in this case is downwards (Fig. 2b) and the Fermi level at the inter-

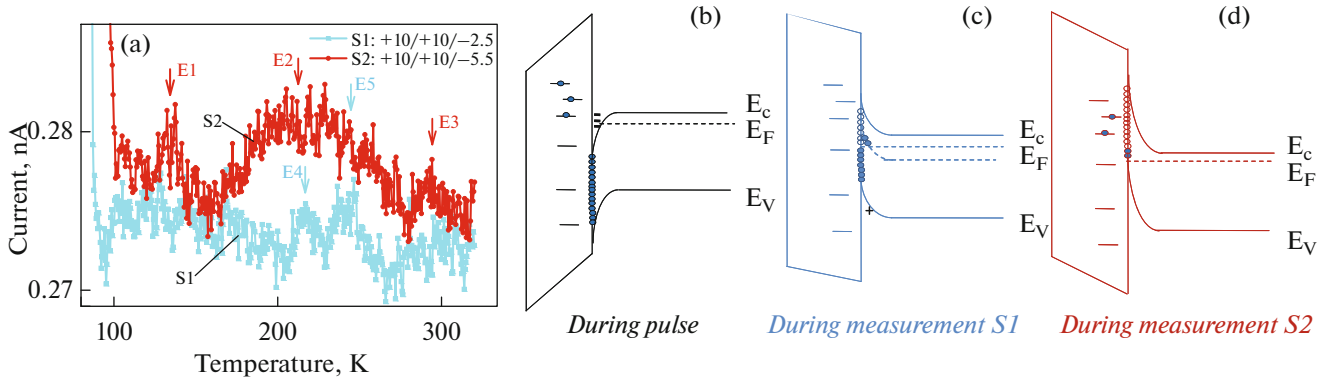


**Fig. 2.** The TSC spectra of e-beam irradiated samples (a) and the band diagrams during the same cooling/pulse conditions (b) and different reverse biases during measurement (c, d). The conditions of the experiment are indicated in the figure.

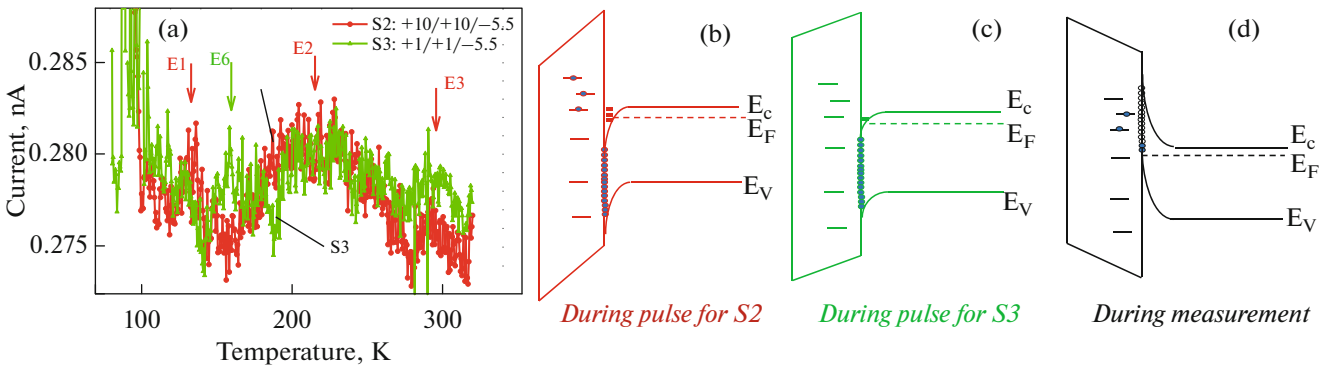


**Fig. 3.** The TSC spectra of e-beam irradiated samples (a) and the band diagrams for different cooling/pulse conditions (b, c) and the same measurement conditions (d). The conditions are indicated in the figure.

face is very close to the conduction band. As a result, the interface states across the bandgap are filled with electrons. The electric field is sufficient to provide an injection of electrons into the oxide bulk via Fowler-Nordheim tunneling [14]. After cooling down to 80 K, a pulse of 10 V for 5 s is applied to maintain the same conditions as during cooling. Afterwards, a reverse bias of −2.5 V was applied and the current was monitored during heating (see TSC spectrum S1 in Fig. 2a). Note that under these bias conditions, the MOS capacitor is driven to a weak depletion. A portion of the bandgap remains filled with electrons (Fig. 2c), and only the interface traps in the upper part of the bandgap are able to contribute to the TSC signal.

In order to identify the origin of the two peaks (labeled E4 and E5) in the TSC spectrum, several additional measurements were carried out using different bias conditions during cooling, pulsing and heating. The second TSC measurement was performed under the same cooling and pulsing conditions, while the magnitude of the reverse bias during heating was increased to −5.5 V (TSC spectrum S2 in Fig. 2a). Under this bias condition during heating the interface states from the whole bandgap (Fig. 2d) contribute to the TSC. As seen from Fig. 2a (curve S2), in the temperature range where peaks E4 and E5 were observed, the thermally stimulated current is notably higher and the TSC spectrum reveals three additional peaks, E1, E2, and E3. The third TSC measurement (spectrum S3 in Fig. 3a) was done under conditions when injection of electrons during cooling and pulsing was eliminated (compare band diagrams in Figs. 3b and 3c), while bias during heating was the same −5.5 V (Fig. 3d).

The peak E1 in Figs. 2a and 3a cannot be related to electron traps in the oxide as it is not seen in the spectrum S1 in Fig. 2a, while the pulsing conditions were favorable for electron injection. We can assume that the E1 peak is the most likely related to electron traps in silicon since it is not seen under the reverse bias −2.5 V (curve S1), when the depletion region width in Si is too narrow as compared to the conditions of the reverse bias of −5.5 V (curve S2). Finally, the peak E1 is very unlikely associated with the interface states due to its temperature position in the TSC spectra. As E1 appears at low (near 130 K) temperature, it should have rather shallow energy level (~0.2 eV below the Si conduction band). However, E1 is not observed under the measurement conditions when the interface states in the upper part of the bandgap should contribute to TSC signal (curve S1).

By comparing three spectra in Figs. 2a and 3a (S1, S2 and S3), the peak E2 can be reliably identified as related to the interface states.

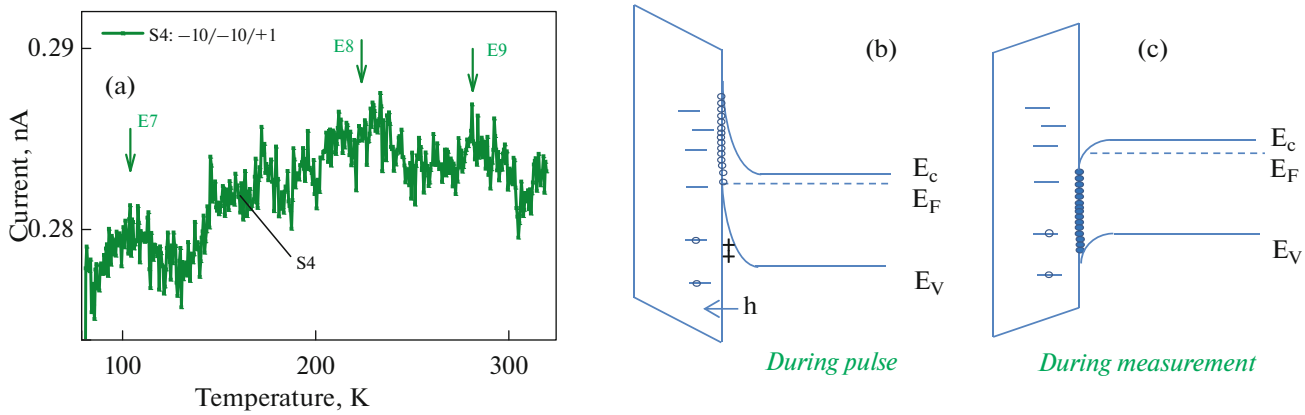


**Fig. 4.** The TSC spectra of e-beam irradiated samples measured under conditions for hole trap detection (a) and the band diagrams for cooling/pulse (b) and measurement (c) conditions.

Peak E3 can be related to the interface states in the lower part of the bandgap, as it is seen in the TSC spectra S2 and S3 in Figs. 2a and 3a, while it is not observed in the S1 spectrum. Taking into consideration the above discussion concerning peak E1, the peak E3 can also be attributed to electron traps in the Si depletion region. For exact identification of the peaks E1 and E3, employment of the deep level transient spectroscopy [15] is required.

The peaks E4 and E5 can be related to the rather deep interface states in an upper part of the bandgap. At the same time, we cannot exclude the possibility for these peaks to be related to electron traps in the oxide bulk. The shallow interface states near the conductance band, which provide significant change in the slope of the C–V curve in a weak accumulation (Fig. 1b), are not seen in the TSC spectra because they appear to be too shallow and release charge at temperature below 80 K.

The peak E6 (spectrum S3 in Fig. 3a) cannot be identified as interface states and/or hole traps in the oxide. We believe it can be related to a very shallow electron trap in the oxide. At a higher magnitude of a volt-

**Table 1.** Identification of the traps revealed in TSC spectra of electron beam irradiated MOS capacitors

| TSC spectrum | Pulse | Bias at measurement | Peaks in TCS spectra | Identification |
|---|---|---|---|---|
| S2 | +10 V (accumulation, electron injection) | −5.5 V (strong depletion; hole injection; all int. states contribute to TSC) | E1, E2, E3 | $E_1$ – electron trap in Si<br>$E_2$ – interface states<br>$E_3$ – interface states in a lower part of the bandgap or electron trap in Si |
| S1 | +10 V (accumulation, electron injection) | −2.5 V (weak depletion; only a portion of the int. states contributes to TSC) | E4, E5 | E4, E5 – electron traps in $SiO_2$ or deep interface states in an upper part of the bandgap |
| S3 | +1 V (accumulation, no electron injection) | −5.5 V (strong depletion; hole injection; all int. states contribute to TSC) | E2, E3, E6 | $E_2$ – interface states<br>$E_3$ – interface states in a lower part of the bandgap or electron trap in Si<br>$E_6$ – shallow electron trap in $SiO_2$ |
| S4 | −10 V (strong depletion, hole injection) | +1 V (accumulation; int. states are filled and do not contribute to TSC) | E7, E8, E9 | Hole traps in $SiO_2$ |

age during cooling and pulsing (curve S2 in Fig. 3a) the electrons can be easily swept away from this trap by high electric field, thus not contributing to the TSC.

The peaks E7, E9 and a broad peak E8 observed in the 200–250 K temperature range (Fig. 4a) are related to hole traps in the oxide bulk. Indeed, the cooling and pulsing conditions for spectrum S4 (Fig. 4a) provided injection of holes into the oxide bulk without any injection of electrons (Fig. 4b). During subsequent heating the MOS capacitor was in a weak accumulation when the interface states remained filled with electrons (Fig. 4c) and did not contribute to TSC.

Table 1 summarizes identification of the peaks observed in the TSC spectra obtained under various cooling, pulsing and heating conditions. One can see that by changing the measurement conditions we are able to identify these traps based on the knowledge of the part of MOS structure that was tested and what carriers were injected and trapped.

## CONCLUSIONS

In this work silicon oxide based, aluminum gated MOS structures have been investigated after irradiation with low energy electrons in scanning electron microscope. The method of thermally stimulated current revealed a number of the e-beam induced charge traps. With the help of the capacitance-voltage method, the traps revealed by the TSC were identified by their location within dielectric, semiconductor or at the interface and by their type as a trap for electrons or for holes.

The TSC method is a powerful tool for investigation of electrically active defects generated in MOS devices (capacitors and field effect transistors) subjected to various kinds of irradiation and technological treatments during either device processing steps or device operation.

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## CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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